



CSE492 ENGINEERING PROJECT PROPOSAL FORM

Date: 29.01.2026

CSE 492 Registration Semester:

Make sure that you have registered to the CSE492 course via OBS before filling out this form.

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<u>CSE 492 registration type (Mark one of the following choices):</u> ☒ Regular ☐ Single Course (Tek Ders) ☐ Supplementary Course (Ek Ders) ☐ Extended	
<u>This is a group project other members are listed:</u> Arda Egemen İPEK Resul ERYURT	
Project Title	YuGoDa: Food Waste Reduction Application (AI Module)
Abstract (200 words)	YuGoDa is an innovative platform created to fight food waste by connecting businesses like restaurants, cafes, and bakeries with customers. The platform allows businesses to sell their surplus food at discounted prices through "Surprise Bags," preventing quality meals from ending up in landfills. Our project is developed by a team of three people, utilizing a Spring Boot backend and a PostgreSQL database to ensure a scalable and reliable system. We have designed management panels for business owners to track their daily inventory and an admin dashboard to oversee the entire ecosystem, including the marketplace operations and financial flow. In this project, I am primarily responsible for the AI-driven assistant and the backend payment infrastructure. Instead of building custom machine learning models, I focus on integrating advanced AI APIs to create a smart recommendation engine. This assistant analyzes user location and order history to provide personalized suggestions through natural language. Users can simply ask, "What are the best vegetarian options near me?" to get instant help. Additionally, I work on the payment logic where the platform collects payments and automatically calculates commissions before sending the remaining share to the restaurants. By combining API-driven intelligence with a secure payment system, I aim to create a seamless experience that helps both the environment and the economy by ensuring no good food goes to waste.
<u>Hardware constraints:(PC, Apple, mobile phone, other device, etc.)</u> - PC/Mac/Linux computers with modern web browsers for accessing the web application. - Development machines: Personal laptops for the 3 people development team.	
<u>Software constraints: (Operating System, Progr. Languages, Libraries, API's, IDE's etc)</u> OS: Cross platform (Windows, macOS, Linux) Languages: JavaScript/TypeScript (React.js frontend), Java 17 (Spring Boot backend). Libraries: React Router, Axios, Redux Toolkit, Spring, Spring Data JPA, LangChain4j. Database: PostgreSQL. APIs: Google Maps API, OpenAI/Claude API (AI chatbot), Firebase Auth (Social Login and Authentication). IDE: VS Code. DevOps: Docker, GitHub. Cloud: AWS.	
<u>Evaluation criteria (Quality Measurements, Scalability, Error Rate, etc)</u> Quality: Minimum %70 unit test coverage, API response time under 500ms. Scalability: Support for 100-200 concurrent users, 3+ restaurant listings. Error Rate: Order processing errors under 10%. Performance: Page load time under 7 seconds, search results under 6 seconds. AI Accuracy: Achieving an F1-Score of 0.75 or higher in recommendation classification; Precision and Recall metrics will be used to evaluate the accuracy of the AI-powered assistant.	

<u>Design Methodology Diagrams</u>	
☒ Use Case Diagrams ☒ Class Diagrams ☒ Sequence Diagrams ☐ State Diagrams	
<u>Impact area</u>	
☒ Health ☒ Environmental ☒ Security ☒ Social / Ethical	
☒ Sustainability ☒ Productivity Other: _____	
Project Advisor Name, Surname, Signature:	Academic Advisor Name, Surname, Signature:
Gürhan KÜÇÜK	Mustafa Bülent MUTLUOĞLU